

DTSA 5517

Fair Machine Learning: Applications

FORMAT	CATEGORY	CREDITS
Online (DTSA)	Elective	1
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 4 – Elective	ML Engineer (76.5 / 100)	71.0 / 100
SPECIALIZATION Fair Machine Learning		

71

CURRENT AVG SCORE
across 6 job roles

100

POTENTIAL SCORE
with recommended additions

+38

MAX SCORE GAIN
best single-role improvement

CURRENT COURSE DESCRIPTION

In this last course, students will deep-dive into previously introduced concepts by implementing these strategies in specific machine learning contexts, such as recommendation systems, generative AI, and representation learning. Technical concepts introduced in the last course will serve as building blocks for more context-specific implementations. Students will also be introduced to specific fairness-related tools commonly used when implementing fairness in practice.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
ML Engineer	76.5	100.0	+23.5	MEDIUM
MLOps Engineer	75.4	100.0	+24.6	MEDIUM
Machine Learning Engineer	73.3	100.0	+26.7	MEDIUM
Applied Scientist	70.5	100.0	+29.5	MEDIUM
AI Engineer	68.0	100.0	+32.0	HIGH
Data Scientist	62.4	100.0	+37.6	HIGH

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (**example**) appear as gaps across 3 or more job roles.

Business Communication

2 of 6 roles

insights

business

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

1

MEDIUM

Consider Business Communication content — relevant to 2 industry roles. Could be introduced via case studies or optional readings. Key terms: insights, business.

GAP KEYWORDS BY ROLE

ML Engineer	76.5 → 100.0 (+23.5 pts)
discovery data models engineering starting build problems impact	
MLOps Engineer	75.4 → 100.0 (+24.6 pts)
detection data text image engineering models	
Machine Learning Engineer	73.3 → 100.0 (+26.7 pts)
data interfaces engineering sets analytics business mobile time	
Applied Scientist	70.5 → 100.0 (+29.5 pts)
discovery scientific research problems models data industry engineering	
AI Engineer	68.0 → 100.0 (+32.0 pts)
following data engineering models build form software problems	
Data Scientist	62.4 → 100.0 (+37.6 pts)
data sets product analytics evolution mitigation insights different	

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.